SyncOS Throughput Analysis

HYPOTHESIS DEFINITION & BASELINE

1. Core Hypotheses

Hypothesis 1: Naive chronological routing drives excessive empty travel

PREMISE: Strict FIFO task assignment forces robots to travel long distances empty, ignoring closer available tasks.

Hypothesis 2: Physical station handovers bottleneck the fleet during shift changes

PREMISE: Throughput drops are caused by human operators or physical production lines delaying material handovers, rather than software routing inefficiencies.

Hypothesis 3: High-density staging areas trigger compounding traffic deadlocks

PREMISE: The mixed fleet naturally converges on shared staging and storage areas, creating spatial traffic jams and non-productive standoff situations.

Hypothesis 4: "Run-to-depletion" charging depletes peak-hour fleet availability

PREMISE: Reactive charging logic pulls robots out of production during peak demand rather than utilizing known idle windows, artificially throttling capacity.

Hypothesis 5: Asymmetric task generation overwhelms specific hardware classes

PREMISE: An imbalance between pallet and intermediate material tasks leaves one hardware class bottlenecked while the other sits underutilized.

Hypothesis 6: Environmental interruptions cause micro-delays that degrade cycle times

PREMISE: Operating in a shared workspace with human workers causes frequent, short-duration pauses that compound into significant 24-hour cycle delays.

2. Data Validation Findings

Based on the analysis of the synthetic operational telemetry (`syncos_task_execution_log_3.csv` and `syncos_fleet_state_log_2.csv`), we have evaluated the hypotheses as follows:

Hypothesis	Metric Calculated	Calculation Method	Value	Status
H1: Naive chronological routing drives excessive empty travel	Empty Travel Share	Average Empty Travel (`Time_Robot_Arrived_Pick` - `Time_Assigned`) / Average Total Cycle Time	`25.7% (119.4s / 465.0s)`	VALIDATED
H2: Physical station handovers bottleneck the fleet during shift changes	Station Wait Time Consistency	Average Wait Time (`Time_Pick_Complete` - `Time_Robot_Arrived_Pick`) grouped across 24 hours	`45.2s avg (range: 43.3s - 47.5s)`	INVALIDATED
H3: High-density staging areas trigger compounding traffic deadlocks	Spatial Concentration of Blockages	% of total `Blocked_Traffic` states grouped by `Location_Zone`	`34.7% Dispatch 26.6% Aisle 1`	VALIDATED
H4: "Run-to-depletion" charging depletes peak-hour fleet availability	Average Charging Threshold	Average `Battery_Level_Pct` at the start of a `Charging` state	`24.0%`	VALIDATED
H5: Asymmetric task generation overwhelms specific hardware classes	Task Volume Distribution	Volume of `Task_ID` grouped by `Task_Type` (Pallet vs Material)	`54.6% Pallet 45.4% Material`	INVALIDATED
H6: Environmental interruptions cause micro-delays that degrade cycle times	Spatial Uniformity of Delays	Distribution of `Blocked_Traffic` events across all facility zones	`Highly concentrated (61%+ in two zones)`	INVALIDATED